

Knitibot Architecture

System Layers

Ingestion

Internal docs & knowledge resources

Kafka Python

Vectorization

Embedding generation

HuggingFace pgvector

Retriever

Semantic search

PostgreSQL pgvector

RAG Engine

Context orchestration

LangChain LLM

API Layer

Secure endpoints

FastAPI OAuth2

Frontend

User Interface

Next.js React

Interactive Diagram

Ingestion

`ingest_pipeline.py` Read-only

```
# ingest_pipeline.py
from kafka import KafkaProducer
import json

producer = KafkaProducer(bootstrap_servers=['localhost:9092'])

def ingest_document(doc):
    """Ingests document into Kafka topic for processing"""
    producer.send('doc-ingest', json.dumps(doc).encode('utf-8'))
    producer.flush()
```

